STRATEGY, LEGAL & OPERATIONS

Production process: Definition, types, and 5 key stages

Learn about the five key stages of the production process and the milestones to track from pre-production through delivery.



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A shipment misses its deadline. Inventory piles up. A quality issue is discovered only after an order is packed. These problems may have different causes, but they often point to the same challenge: a

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Production processes vary by product, volume, and level of customization. A craft business may make one item at a time, while a high-volume manufacturer may use an assembly line. Whatever the model, effective production depends on coordinating materials, labor, equipment, quality, costs, and deadlines.

This guide explains what the production process is, the main production methods, five key stages used to plan and control production, and the milestones and metrics manufacturers can track to improve performance.

Key takeaways

- A production process turns inputs such as raw materials, labor, equipment, and other resources into finished goods or services.
- Common production methods include mass, batch, job, service, craft, and mass customization, with the right approach depending on volume, variation, capacity, and customer requirements.
- Five key production process stages are planning, routing, scheduling, dispatching, and follow-up, which help manufacturers coordinate work and compare actual performance with the plan.
- Tracking milestones from pre-production through delivery can help teams identify quality, scheduling, and fulfillment issues earlier, while production metrics and connected data support ongoing improvement.

Here's what we'll cover

- What is the production process?
- What are the main types of production processes?
- What are the five stages of the production process?
- Which production milestones should you track?
- How can software improve the production process?
- Frequently asked questions about the manufacturing production process

What is the production process?

The production process is the series of activities a business uses to turn inputs—such as raw materials, labor, equipment, and other resources—into finished goods or services. In manufacturing, this can include preparing materials, completing production operations, carrying out quality checks, and preparing finished goods for release.

The exact production process depends on what the business produces, the required volume, and how much variation or customization customers expect. Although the details vary, each stage moves materials or work toward the finished product and helps manufacturers coordinate resources, quality, costs, and delivery requirements..

What are the main types of production processes?

Production processes in manufacturing can take several forms. Common production methods include

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Production type	Best suited to	Example
Mass production	High volumes of standardized or identical products.	An assembly line producing standardized products at high volume.
Batch production	Related products made in separate groups or production runs.	One product produced in different colors, sizes, or pack formats.
Job production	One-off or highly customized products made individually or in small quantities.	Shipbuilding or custom machinery.
Service production	Processes designed to deliver a service rather than manufacture a physical product.	Banking, IT support, or food preparation.
Craft production	One product or a small number of handmade items.	Custom dressmaking.
Mass customization	High production volumes with selected customer-specific options.	Products tailored by size, configuration, or finish.

These production methods aren't always mutually exclusive. For example, a manufacturer might mass-produce standard components, use batch production for product variations, and apply craft methods for final customization.

A supply chain diagram can help you visualize how materials, information, and finished goods move through the wider process.

What are the five stages of the production process?

The five stages of the production process are planning, routing, scheduling, dispatching, and follow-up. Together, they provide a framework for deciding what needs to be produced, how and when the work will be completed, and how performance will be monitored against the plan.

The exact manufacturing process steps vary by product and industry. However, these five stages help manufacturers coordinate resources, production activities, deadlines, and performance from initial planning through to completion.

1. Planning

Production planning defines what needs to be produced and the resources required to do it. The plan sets out the work involved, who is responsible, and what resources need to be available before work begins.

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much time, equipment, and production capacity each order will require.

Production planning doesn't always require a complex system. Smaller businesses can begin with an Excel production planning template. As production volume and complexity increase, manufacturers may benefit from production management software that brings planning, scheduling, and production tracking in one connected system.

2. Routing

Production routing defines the sequence of operations required to make a product and where each operation will take place. The route may identify machines, production lines, work centers, or departments involved at each stage.

A production route may also specify expected setup and processing times for each operation. These details help manufacturers understand resource requirements and provide the foundation for an achievable production schedule.

3. Scheduling

Production scheduling determines when each job, operation, or batch should start and finish. It combines routing information with factors such as batch sizes, setup requirements, available production capacity, and customer delivery dates.

A realistic schedule helps manufacturers sequence work, coordinate labor, materials, and equipment, and spot potential capacity problems before they cause delays. Sage manufacturing documentation similarly describes scheduling as setting production dates based on successive operations and their resources.

4. Dispatching

Production dispatching releases scheduled work so production can begin. This may involve issuing work orders and instructions and making the required materials and tooling available to the production team.

Once work is underway, manufacturers can record progress through each operation. For example, barcode scanning or other shop-floor tracking methods can show when work enters and leaves an operation, giving teams greater visibility into the status of an order.

5. Follow-up and production control

Production follow-up, sometimes referred to as production control, compares actual performance with the production plan. Teams can use this comparison to identify delays, quality problems, cost variances, and other differences between expected and actual performance.

Where problems are identified while production is still underway, teams can take corrective action. Reviewing recurring variances can also improve future plans and reveal opportunities to reduce bottlenecks, control costs, or improve wider production performance.

Which production milestones should you track?

A useful way to keep orders on track is to monitor four production milestones: pre-production, production, pre-shipment, and shipment through delivery.

These checkpoints cover the order lifecycle and help teams identify quality, quantity, and scheduling problems before they affect the customer.

Reviewing performance only after an order is complete can improve future production plans, but it can't prevent problems with the current order. Tracking milestones while work is in progress gives teams an opportunity to correct issues before they lead to rework, delays, excess costs, or unsellable

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Production milestone	What to check	Practical example
Pre-production	Confirm that materials meet the required standard, colors match where relevant, packaging meets specifications, and equipment or tooling is ready. Equipment should be calibrated where required, and machine settings configured to meet the product specifications.	A personal care brand may approve color and packaging samples before production. New packaging tooling should then be inspected and signed off on.
Production	Compare progress with the schedule and monitor quality throughout the manufacturing process. Early checks can reduce rework, scrap, and production delays.	A clothing manufacturer may track cutting, sewing, finishing, and packing tasks. Teams can use a production tracker to assign responsibilities, define checkpoints, and give internal teams and suppliers clear instructions.
Pre-shipment	Inspect the finished product and packaging, confirm that customer requirements have been fulfilled, and collect any documentation needed before release.	Depending on the product and market, a food or beverage company may request certificates of analysis, quality release forms, or third-party testing documents before shipment.
Shipment and delivery	Confirm that the correct product and quantity are sent to the right destination and meet the required quality and delivery requirements. Customers should also receive accurate shipping and tracking updates.	Teams can cross-reference each order with its documentation at every handoff. They should also confirm that tracking information reflects the shipment's actual status rather than showing that a label is still being created after the goods have left the facility.

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Checklists and production-tracking templates can make milestone reviews more consistent and reduce the chance that an important requirement is missed. Clear hand-offs between production, fulfillment, and shipping teams can also help maintain accurate records, communicate delays, and give customers dependable delivery information.

How can software improve the production process?

Production management software can improve the production process by bringing schedules, work orders, inventory information, quality data, and production costs into one connected system. This gives teams greater visibility into work in progress and can help them identify delays, material shortages, quality issues, or cost variances sooner.

As operations become more complex, integrated ERP and production management software can also connect production with purchasing, inventory, supply chain, and finance, reducing manual handoffs between teams and systems. The right system should support the business's production method, operating scale, planning requirements, and reporting needs.

supply chain software can reduce manual follow-up by automating routine checks, organizing documents, and comparing planned activity with actual results. The right system should support the business's production method, operating scale, and reporting needs.

Final thoughts

A well-managed production process helps manufacturers coordinate resources, control costs and quality, and keep orders moving toward delivery. Clear planning, regular performance checks, and connected production data can help teams identify problems earlier and make better production decisions over time.

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Frequently asked questions about the manufacturing production process

How do you choose the right type of production process?

Choose a production process based on production volume, product variation or customization, available capacity, and customer requirements. Businesses should also consider labor and material costs, inventory needs, available technology, profit margins, and the capital investment required to carry out the process.

What is the difference between a production process and a manufacturing process?

A production process is the broader set of activities used to turn inputs into goods or services. A manufacturing process specifically concerns making physical goods from materials or components, so manufacturing is a type of production rather than an exact synonym.

Which metrics help measure production process performance?

Useful production metrics include cycle time, throughput, on-time delivery, equipment downtime, production yield, and scrap or rework rates. These metrics help manufacturers understand production speed, reliability, quality, and efficiency.

Throughput measures how much is produced in a set period, while production yield shows how much usable output the process creates. Manufacturers can also compare planned costs, quantities, and

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Manufacturers can improve their production process by mapping the current workflow, identifying bottlenecks, and comparing actual results with the production plan. Standardizing repeatable tasks, clarifying responsibilities, training employees, and automating routine data collection can also help reduce delays and errors.

Manufacturers looking to connect automation, equipment, production data, and business systems can also explore how a smart factory supports more coordinated production decisions.

What is an example of a production process?

One example of a production process is a furniture manufacturer turning timber and other components into finished products. Depending on the product, the manufacturing process might include cutting and machining materials, assembly, finishing, quality checks, and packaging.

Planning, routing, and scheduling help coordinate the labor, materials, equipment, and timing required to complete those manufacturing steps efficiently.

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